

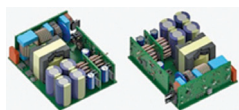
8GB eMMC, and a 32kbit EEPROM. It supports various interfaces, including UART, CAN FD, Gigabit Ethernet, USB, and more. A development board for prototyping includes the SOM and expansion for additional peripherals like Wi-Fi, Bluetooth, 4G/5G, and display interfaces.

MYIR

<https://www.myirtech.com>

PD evaluation board

Eggtronic's SmartEgg 240W PD 3.1 evaluation board (EVB) streamlines the development of high-power 240W power delivery (PD) 3.1 applications,



reducing both size and cost while boosting efficiency.

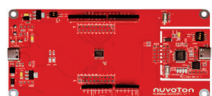
It achieves over 95% peak efficiency and maintains over 90% across the load range, using a single-stage design with zero voltage switching (ZVS), power factor correction (PFC), and quasi-forward isolated regulation. This approach cuts the bill of materials and component size, offering up to 50% energy savings at light loads and 21.9W/in³ power density.

Eggtronic

<https://www.eggtronic.com>

32-bit microcontroller

Advancements in semiconductor technology have enhanced the performance and reduced the cost of 32-bit microcontrollers, driving their widespread use in applications like



AIoT, industrial automation, smart homes, energy storage, and automotive electronics. Nuvoton Technology addresses these needs with the M2003 series, based on the ARM Cortex-M23 core, offering features like a hardware multiplier/divider, 32kB Flash, and 4kB SRAM. Integrated peripherals include timers, UARTs, PC, ADC, and PWM. The series supports ISP/IAP updates and

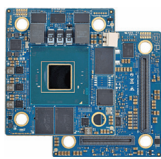
provides a complete development ecosystem with NuTool – CodeGenerator, Keil, IAR, and NuEclipse IDE support.

Nuvoton Technology Corporation

<http://www.nuvoton.com>

System-on-module

iWave recently introduced the iW-RainboW-G58M system-on-module (SoM) featuring the Intel Agilex 5



FPGA series, expanding its FPGA offerings for midrange applications. Supporting Intel Agilex 5 E-Series SoCs and the B32A package, this compact SoM (60mm x 70mm) operates in harsh conditions (-40°C to +85°C) and offers long-term availability with a 15-year longevity program.

iWave

<https://www.iwavesystems.com>

MANUFACTURING EQUIPMENT

High-speed chip moulder

Hanwha Precision Machinery's XM520, a general-purpose chip moulder, is redefining surface mount technology (SMT) and through-hole technology (THT) production. With high-speed capabilities, it can place up to 100,000 components per hour with $\pm 22\mu\text{m}$



accuracy. It supports component sizes from 0201 chips to 22mm, and PCBs up to 150mm x 55mm. Awarded the 2024 NPI Award for high-speed component placement, the XM520 features advanced tools like the stage camera and docking cart for seamless operation. Its compact design, dual gantry, and 10-spindle configuration offer exceptional flexibility, making it ideal for diverse production environments.

Hanwha Precision Machinery

<https://www.hanwha-pm.com/en>

TEST & MEASUREMENT

Intelligent circuit board detection system

JB Instruments introduces the Intelligent circuit board detection system, a solution for PCB testing. This system optimises testing by selecting the most



suitable method for each PCB's unique characteristics, ensuring comprehensive and efficient inspections. Central to its functionality

is the JBM-SE smart camera, featuring high-precision positioning, measurement calibration, advanced OCR, and a multi-focal-length lens. With built-in multi-colour lighting, RS-232, and Ethernet outputs, it integrates seamlessly into various processes. The system excels in detecting, measuring, and verifying patterns, edges, and colours, with advanced tools for logic judgment, defect detection, and precise calibration, ensuring reliable PCB inspections.

JB Instruments

<https://www.jbinstruments.com>

A huge neutrino detector

Neutrinos, once thought to be massless, are now believed to have a small mass and can oscillate between three 'flavours.' These elusive particles pass through matter undetected, making them hard to study. To address this, the Euro 400 million JUNO detector, located 750 metres underground



in Jiangmen, China, was created by 730 scientists from 17 countries. Operational by 2024, JUNO will be the world's largest neutrino detector, advancing particle physics and multi-messenger astronomy.

Spectrum Instrumentation

<https://spectrum-instrumentation.com>

Optical emission spectrometer

Metal Power Analytical Pvt Ltd (MPA), an Indian manufacturer of analytical